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### Method for securing brake fluid

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#### Abstract

A method for controlling a brake system in the case of leaks is provided. The brake system has a mechanically actuable master brake cylinder, which is connected to a first partial reservoir of a brake fluid vessel, and has an electric linear actuator, which is connected to a second partial reservoir of the brake fluid vessel. The master brake cylinder and the linear actuator are connected via an electrically closable circuit isolating valve. In order to protect the brake fluid, in the event of a leak during a transition to a standby state, the linear actuator is activated to take a brake fluid volume out of the first partial reservoir via the open circuit isolating valve and to displace it into the second partial reservoir.

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## References Cited

### U.S. PATENT DOCUMENTS

| Patent No.   | Issued Date | Patentee Name | U.S. Cl. | CPC         |
|--------------|-------------|---------------|----------|-------------|
| 4125234      | 12/1977     | Tregre        | 188/351  | B60T 13/16  |
| 2011/0175436 | 12/2010     | Nakata        | 303/6.01 | B60T 8/4081 |
| 2015/0035351 | 12/2014     | Okano         | N/A      | N/A         |
| 2016/0032995 | 12/2015     | Nishino       | 188/72.3 | F16D 65/18  |
| 2020/0307536 | 12/2019     | Biller        | N/A      | N/A         |
| 2021/0339728 | 12/2020     | Gaughan       | N/A      | B60T 13/665 |
| 2022/0169249 | 12/2021     | Gaughan       | N/A      | B60T 17/22  |
| 2022/0324430 | 12/2021     | Watanabe      | N/A      | B60T 11/26  |
| 2023/0202447 | 12/2022     | Lee           | 200/82D  | B60T 13/146 |
| 2023/0234545 | 12/2022     | Yuan          | N/A      | F16D 65/18  |
| 2023/0286488 | 12/2022     | Kim           | N/A      | B60T 13/686 |

### FOREIGN PATENT DOCUMENTS

| Patent No.   | Application Date | Country | CPC |
|--------------|------------------|---------|-----|
| 102011082492 | 12/2012          | DE      | N/A |
| 112012005989 | 12/2013          | DE      | N/A |
| 10343985     | 12/2016          | DE      | N/A |
| 102016222578 | 12/2017          | DE      | N/A |
| 102017222440 | 12/2017          | DE      | N/A |
| 102017222789 | 12/2018          | DE      | N/A |
| 102018211088 | 12/2019          | DE      | N/A |
| 102019213057 | 12/2020          | DE      | N/A |
| 102019219158 | 12/2020          | DE      | N/A |
| 3724045      | 12/2019          | EP      | N/A |

### OTHER PUBLICATIONS

Japanese Office Action dated Feb. 7, 2025 in correspondence with Japanese Application No. 2023-574205. cited by applicant  
Korean Office Action dated Jan. 23, 2025 in correspondence with Korean Application No. 10-2023-7039121. cited by applicant  
German Search Report dated Apr. 14, 2022 for the priority German Patent Application No. 10 2021 206 182.6. cited by applicant  
The International Search Report and the Written Opinion of the International Searching Authority mailed on Oct. 31, 2022 for the PCT Application No. PCT/DE2022/200107. cited by applicant

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## **Background/Summary**

### TECHNICAL FIELD

(1) The embodiments relate to a method for controlling a brake system in the case of leaks.

### BACKGROUND

(2) When initializing a brake system, for example when unlocking the vehicle or when a driver is approaching the vehicle, the filling level of the brake fluid vessel is typically checked. If this filling level lies below a threshold, it is assumed that there is a leak in the braking system.

(3) The degradation of the brake system in the event of an external leak is the implementation of circuit separation. For this purpose, the single-circuit by-wire system is separated into a two-circuit system consisting of a driver's circuit with the master brake cylinder and a plunger circuit with the linear actuator by way of the circuit isolating valve being energized.

(4) In the de-energized case, that is in particular in a standby state, when the vehicle is switched off, the brake system remains a single-circuit system since the circuit isolating valve has to be energized for circuit separation. As a result, in the case of a leak in the direction of one of the four wheels, the brake fluid volume of the first partial chamber, which is connected to the master brake cylinder, is lost.

(5) In order to minimize the loss of brake fluid, circuit separation that has already been completed is often maintained for, for example, 48 hours in an extended run-on period with the ignition switched off and the circuit valve is energized. In addition, after waking up from a sleep mode and a plunger circuit leak being identified, the first partial chamber is refilled with brake fluid volume from the second partial chamber.

(6) However, these two functions are subject to restrictions. The extended run-on period is limited in time. At the latest after the end of the run-on period, there is a transition to a single-circuit braking system and thus a loss of the volume of the first partial chamber. The duration of the extended run-on period depends on the charge level of the battery. The extended run-on period does not completely protect against loss of the volume of the first partial chamber either. Gravitational pressure generates a leakage flow via the closed circuit isolating valve, which can lead to emptying of the partial chamber. Refilling the first partial chamber reduces the brake fluid volume of the second partial chamber by the corresponding amount. Refilling is therefore limited.

### SUMMARY

(7) The object is therefore to increase the readiness of the brake system in the case of a leak.

(8) A method in the event of a leak during a transition to a standby state is provided. A linear actuator is activated to take a brake fluid volume out of the first partial reservoir via the open circuit isolating valve and to displace it into the second partial reservoir. The brake fluid is thus protected from leaking even in the sleep mode without the valves being energized.

(9) In one embodiment, when a leak is detected, the brake system is separated into two partial circuits by closing the circuit isolating valve, wherein a first partial circuit comprises at least the master brake cylinder with the first partial reservoir and also two first wheel brakes with associated inlet valves and outlet valves, and a second partial circuit comprises at least the linear actuator with the second partial reservoir and also two second wheel brakes with associated inlet valves and outlet valves. The displacement of the brake fluid volume from the first into the second partial chamber can then be performed for example before the circuit separation, for example in the sleep mode, is canceled.

(10) In another embodiment, the brake fluid volume is displaced from the first partial chamber into the second partial chamber only if the leak is present in the second partial circuit. For example, the method can also be implemented specifically when at least one wheel brake in the second partial circuit has a leak and at least one wheel brake does not have a leak.

(11) In another embodiment, the circuit separation is maintained for a predetermined period of time during a transition to a standby state, wherein the brake fluid volume of the first partial chamber is displaced into the second partial chamber at the end of the predetermined period of time. This allows the brake fluid to be protected even when the state of charge of the battery is low.

(12) In another embodiment, the brake system is designed in such a way that, in a de-energized state, the first partial chamber is connected to the wheel brakes. Such a brake system has the advantage that the brake system can also be operated without electricity by pure hydraulics.

(13) In another embodiment, a brake fluid volume is displaced from the first partial reservoir into the second partial reservoir until the first partial reservoir is emptied. This can be monitored and controlled by a filling level sensor of the first partial chamber or brake fluid volume is displaced until emptying can be assumed given the existing size ratios of the individual vessels.

(14) In another embodiment, for the purpose of displacing the brake fluid volume out of the linear actuator into the second partial reservoir, the circuit isolating valve is closed, the inlet valve of the wheel brake with a leak is closed, the inlet valve and the outlet valve of the wheel brake without a leak are opened and the linear actuator is operated in the pressure build-up direction. The brake fluid volume is thus conveyed into the brake fluid vessel via the wheel without a leak. In order to determine which wheel brake has the leak, for example, the linear actuator can be connected with open flow individually to the wheel brakes in succession and a hydraulic pressure can be built up by the linear actuator. In so doing, a check is made as to what distance is required to build up the pressure in each case, as a result of which it is identified whether there is a leak at the respective wheel brake. The wheel brake at which the leak is located is thus identified.

(15) In another embodiment, for the purpose of taking the brake fluid volume out of the first partial chamber into the linear actuator, an open-flow connection is established between the linear actuator and the first partial chamber and the linear actuator is operated in the pressure reduction direction. Brake fluid is thus drawn from the first partial chamber.

(16) In another embodiment, the open-flow connection is established via the master brake cylinder and/or via inlet valves and outlet valves of at least one wheel brake. In the process, suction is preferably performed via the master brake cylinder since this prevents brake fluid from being conveyed backwards through inlet valves and outlet valves. This is risky due to contamination.

(17) In another embodiment, when the brake fluid volume is taken out of the first partial chamber, the linear actuator is operated in such a way that a generated vacuum remains lower than a threshold value.

(18) In another embodiment, the linear actuator is connected to the second partial chamber via a non-return valve, which opens at a return differential pressure, and wherein the threshold value is lower than twice the return differential pressure. This prevents the linear actuator from sucking brake fluid primarily out of the second partial chamber. The threshold value may be less than 150%, for example less than 100%, of the return differential pressure.

(19) The object is also achieved by a brake system for a motor vehicle comprising a mechanically

actuatable master brake cylinder, which is connected to a first partial reservoir of a brake fluid vessel, and an electric linear actuator, which is connected to a second partial reservoir of the brake fluid vessel, wherein the master brake cylinder and the linear actuator are connected via an electrically closable circuit isolating valve, wherein a control unit of the brake system is designed to carry out one of the preceding methods.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Further features, advantages and possible applications can also be found in the description below of exemplary embodiments and the drawings. All of the features described and/or pictorially depicted belong to the subject matter of the invention both individually and in any combination, also independently of their summarization in the claims or the back-references thereof.
- (2) FIG. 1 schematically shows an exemplary brake system,
- (3) FIG. 2 shows a graph of the brake fluid volume of the first partial chamber,
- (4) FIG. 3 shows a graph of the brake fluid volume of the second partial chamber,
- (5) FIG. 4 shows a graph of the brake fluid volume of the first partial chamber,
- (6) FIG. 5 shows a graph of the brake fluid volume of the second partial chamber,
- (7) FIG. 6 shows a flowchart comprising one embodiment of a method of operating a brake system.

### DETAILED DESCRIPTION

- (8) A braking system for a motor vehicle shown in FIG. 1 comprises four hydraulically actuatable wheel brakes **8a-8d**. The brake system comprises a master brake cylinder **2** which can be actuated by means of an actuating or brake pedal **1**, a travel simulator or a simulation device **3** which interacts with the master brake cylinder **2**, a pressure medium storage tank **4** which is under atmospheric pressure, an electrically controllable pressure application device **5**, and wheel-specific brake pressure modulation valves which are configured according to the example as inlet valves **6a-6d** and outlet valves **7a-7d**. Furthermore, the braking system comprises an electronic open-loop and closed-loop control unit **12** for controlling the electrically actuatable components of the brake system. This can also be composed of several individual control devices.
- (9) According to the example, the wheel brake **8a** is assigned to the left front wheel (FL), the wheel brake **8b** is assigned to the right front wheel (FR), the wheel brake **8c** is assigned to the left rear wheel (RL), and the wheel brake **8d** is assigned to the right rear wheel (RR).
- (10) The master brake cylinder **2** has, in a housing **16**, a master brake cylinder piston **15**, which delimits a hydraulic pressure chamber **17**, and constitutes a single-circuit master brake cylinder. The pressure chamber **17** receives a restoring spring **9** which, with the master brake cylinder **2** unactuated, positions the piston **15** in a starting position. At one end, the pressure chamber **17** is connected to the pressure medium storage tank **4** via radial bores, which are formed in the piston **15**, and a corresponding pressure equalization line **41**, wherein this connection can be shut off by a relative movement of the piston **17** in the housing **16**. At the other end, the pressure chamber **17** is connected by means of a hydraulic line section (also referred to as first feed line) **22** to a brake supply line **13** to which the input connections of the inlet valves **6a-6d** are connected. The pressure chamber **17** of the master brake cylinder **2** is thus connected to all inlet valves **6a-6d**.
- (11) According to the example, no valve, for example no electrically or hydraulically actuatable valve and no non-return valve, is arranged in the pressure equalization line **41** or in the connection between the pressure chamber **17** and the pressure medium storage tank **4**.
- (12) As an alternative, a normally open, diagnostic valve, for example a connection in parallel between a normally open diagnostic valve and a non-return valve which closes in the direction of the pressure medium storage tank **4**, can be contained in the pressure equalization line **41** or between the master brake cylinder **2** and the pressure medium storage tank **4**.

(13) An isolating valve **23** is arranged between the feed line **22**, which is connected to the pressure chamber **17**, and the brake supply line **13**, or the pressure chamber **17** is connected to the brake supply line **13** via the first feed line **22** with an isolating valve **23**. The isolating valve **23** is designed as an electrically actuatable, for example normally open (NO), 2/2-way valve. The hydraulic connection between the pressure chamber **17** and the brake supply line **13** can be shut off by the isolating valve **23**.

(14) A piston rod **24** couples the pivoting movement of the brake pedal **1** as a result of pedal actuation to the translational movement of the master brake cylinder piston **15**, the actuation travel of which is detected by a travel sensor **25** which may be of redundant design. In this way, the corresponding piston travel signal is a measure of the brake pedal actuation angle. It represents a braking demand of a vehicle driver.

(15) A pressure sensor **20**, which is connected to the first feed line **22**, detects the pressure built up in the pressure chamber **17** as a result of a displacement of the piston **15**. This pressure value can also be evaluated to characterize or determine the braking demand of the vehicle driver. As an alternative to a pressure sensor **20**, use can also be made of a force sensor **20** for determining the braking demand of the vehicle driver.

(16) According to the example, the simulation device **3** is of hydraulic configuration and is hydraulically coupled to the master brake cylinder **2**. The simulation device **3** substantially has, for example, a simulator chamber **29**, a simulator rear chamber **30** and a simulator piston **31** which separates the two chambers **29**, **30** from each other. The simulator piston **31** is supported on a housing by an elastic element **33** (e.g. simulator spring) which is arranged in the simulator rear chamber **30** (which is dry according to the example). According to the example, the hydraulic simulator chamber **29** is connected to the pressure chamber **17** of the master brake cylinder **2** by means of a preferably electrically actuatable, for example normally closed simulator enable valve **32**.

(17) The braking system or the brake system comprises an inlet valve **6a-6d** and an outlet valve **7a-7d** for each hydraulically actuatable wheel brake **8a-8d**, the inlet valves and outlet valves being hydraulically interconnected in pairs via center connections and connected to the wheel brake **8a-8d**. A non-return valve, not specifically designated, which opens in the direction of the brake supply line **13**, is connected in parallel to each of the inlet valves **6a-6d**. The output connections of the outlet valves **7a-7d** are connected to the pressure medium storage tank **4** via a common return line **14**.

(18) The electrically controllable pressure-providing device **5** is designed as a hydraulic cylinder-piston arrangement (or as a single-circuit, electrohydraulic actuator (linear actuator)), the piston **36** of which can be actuated by a schematically indicated electric motor **35** with interconnection of a likewise schematically illustrated rotation-translation mechanism **39**. The piston **36** delimits the single pressure chamber **37** of the pressure-providing device **5**. A merely schematically indicated rotor position sensor which serves to detect the rotor position of the electric motor **35** is denoted by reference sign **44**.

(19) A line section (also referred to as second feed line) **38** is connected to the pressure chamber **37** of the electrically controllable pressure-providing device **5**. The feed line **38** is connected to the brake supply line **13** via an electrically actuatable, normally closed, sequence valve **26**. The sequence valve **26** allows the hydraulic connection between the pressure chamber **37** of the electrically controllable pressure-providing device **5** and the brake supply line **13** (and thus the input connections of the inlet valves **6a-6d**) to be opened and shut off in a controlled manner. The actuator pressure produced by the action of force of the piston **36** on the pressure medium enclosed in the pressure chamber **37** is fed into the second feed line **38**. In a "brake-by-wire" mode, for example in a fault-free state of the brake system, the feed line **38** is connected to the brake supply line **13** via the sequence valve **26**. In this way, there is, during normal braking, a build up and a reduction in wheel brake pressure for all the wheel brakes **8a-8d** owing to the forward and backward movement of the piston **36**.

(20) In the case of a reduction in pressure by backward movement of the piston **36**, the pressure medium previously displaced from the pressure chamber **37** of the pressure-providing device **5** into the wheel brakes **8a-8d** flows back again into the pressure chamber **37** in the same way.

(21) As an alternative, wheel brake pressures which differ in a wheel-specific manner can be easily adjusted by means of the inlet and outlet valves **6a-6d**, **7a-7d**. In the case of a corresponding reduction in pressure, the portion of pressure medium discharged via the outlet valves **7a-7d** flows via the return line **14** into the pressure medium storage tank **4**. Additional pressure medium can be drawn into the pressure chamber **37** owing to a backward movement of the piston **36** with the sequence valve **26** closed by way of pressure medium being able to flow out of the tank **4** into the actuator pressure chamber or pressure chamber **37** via the line **42** with a non-return valve **45**, which opens in a flow direction to the actuator **5**. The non-return valve **45** typically opens starting at a differential pressure of approximately 0.08 bar.

(22) According to the example, the pressure chamber **37** is also connected to the pressure medium storage tank **4** via one or more snifter holes in an unactuated state of the piston **36**. This connection between the pressure chamber **37** and the pressure medium storage tank **4** is disconnected when the piston **36** is (sufficiently) actuated in the actuating direction.

(23) In the brake supply line **13** there is arranged an electrically actuatable, normally open circuit isolating valve **40** by means of which the brake supply line **13** can be separated into a first line section **13a**, which is connected (via the isolating valve **23**) to the master brake cylinder **2**, and a second line section **13b**, which is connected (via the sequence valve **26**) to the pressure-providing device **5**. The first line section **13a** is connected to the inlet valves **6a**, **6b** of the wheel brakes **8a**, **8b**, and the second line section **13b** is connected to the inlet valves **6c**, **6d** of the wheel brakes **8c**, **8d**.

(24) With the circuit isolating valve **40** open, the brake system is of single-circuit design. The brake system can be separated or divided into two brake circuits (partial circuits) I and II, in for example controlled according to the situation, by closing the circuit isolating valve **40**. Here, in the first brake circuit I, the master brake cylinder **2** is connected (via the isolating valve **23**) to only the inlet valves **6a**, **6b** of the wheel brakes **8a**, **8b** of the front axle VA, and, in the second brake circuit II, the pressure-providing device **5** is connected (with the sequence valve **26** open) to only the wheel brakes **8c** and **8d** of the rear axle HA.

(25) With the circuit isolating valve **40** open, the input connections of all the inlet valves **6a-6d** can be supplied by means of the brake supply line **13** with a pressure which corresponds to the brake pressure which is provided by the pressure-providing device **5** in a first operating mode (e.g. "brake-by-wire" operating mode). In a second operating mode (e.g. in a de-energized fallback operating mode), the pressure of the pressure chamber **17** of the master brake cylinder **2** can be applied to the brake supply line **13**.

(26) The brake system comprises a level-measuring device **50** for determining a pressure medium level/filling level in the pressure medium storage tank **4**.

(27) The pressure medium storage tank **4** has a first partial chamber **10** and a second partial chamber **11**, which are separated from each other via a bulkhead. The first partial chamber **10** provides brake fluid for the master brake cylinder **2** via the pressure equalization line **41**. The second chamber **11**, on the other hand, supplies the linear actuator **5** with brake fluid via the pressure equalization line **42** and the non-return valve **120**.

(28) According to the example, the hydraulic components, namely the master brake cylinder **2**, the simulation device **3**, the pressure-providing device **5**, the valves **6a-6d**, **7a-7d**, **23**, **26**, **40** and **32** and also the hydraulic connections including the brake supply line **13**, are arranged together in a hydraulic open-loop and closed-loop control unit **60** (HCU). The electronic open-loop and closed-loop control unit (ECU) **12** is assigned to the hydraulic open-loop and closed-loop control unit **60**. The hydraulic and the electronic open-loop and closed-loop control unit **60**, **12** are configured as one unit (HECU).

(29) The brake system comprises a pressure sensor **19** or system pressure sensor for detecting the pressure provided by the pressure-providing device **5**. Here, the pressure sensor **19** is arranged behind the sequence valve **26** as seen from the pressure chamber **37** of the pressure-providing device **5**.

(30) If pressure is reduced over a relatively long period of time or several times in succession via the outlet valves **7a** to **7d** of the wheel brakes **8a** to **8d**, the piston **36** of the linear actuator **5** moves successively forward until it has reached its foremost position, from which an additional volume flow can no longer be provided. At this point in time at the latest, the linear actuator has to be refilled by means of a refill.

(31) For such a refill of the linear actuator **5**, the sequence valve **26** is closed and then the piston **36** is moved back. As a result, brake fluid is drawn out of the reservoir **4** via the non-return valve **45**.

(32) When initializing the brake system, for example when unlocking the vehicle, the level-measuring device **50** is used to check whether the brake fluid level is sufficient. If this is the case, the brake system starts in the normal operating mode. If, on the other hand, the brake fluid level lies below a threshold value, the single-circuit brake system is converted into a two-circuit brake system by closing the circuit isolating valve **40**. As a result, the partial circuit not affected by the leak is effectively protected from the leak.

(33) In addition, a leak test is typically performed to locate the leak. For this purpose, the linear actuator **5** is connected successively via the sequence valve **26** and an open inlet valve **6** to the individual wheel brakes **8**, the outlet valve **7** of which is closed. The remaining inlet valves **6** are each closed here. A pressure is built up here by means of the linear actuator **5** and the required travel is controlled. If the travel is abnormally high or the pressure cannot be maintained, the brake system assumes a leak at the corresponding wheel.

(34) When the braking system enters a sleep mode after the vehicle is parked, all valves generally move to their de-energized position. Since the circuit isolating valve **40** is configured as a normally open valve, circuit separation is lost in the sleep mode. In order to prevent this, the circuit separation can still be maintained in the sleep mode for a specified period of time in a so-called run-on period. However, since the energy reserves are limited by the vehicle battery, this extended run-on period is also limited.

(35) FIG. 2 to FIG. 5 show time profiles of the volumes of the first partial chamber **10** and the second partial chamber **11** for a situation in which the vehicle is repeatedly operated with circuit separation for one hour and is then parked in the sleep mode for 23 h.

(36) FIG. 2 and FIG. 3 show the time profiles from the method known from the prior art. A leak has already been detected at the rear-axle wheel brake **8d** and the active circuit separation has been implemented by closing the circuit isolating valve **40**. In addition, the sequence valve **26** has been closed and the linear actuator **5** has been switched off. The brake system is accordingly braked by the muscle power of the driver and the master brake cylinder **2** on the front wheel brakes **8a** and **8b**. As shown in FIG. 2, the filling level of the first partial chamber **10** is just below 20 cm.sup.3 at the beginning, this corresponding to a full first partial chamber **10**. As a result of the leak, some brake fluid volume is now slowly lost during operation via the closed circuit isolating valve **40** since this does not close completely tightly. The volume of the second partial chamber **11** remains constant, protected by the closed sequence valve **26**, as can be seen from the graph of FIG. 3. When the vehicle is turned off, the brake system enters a de-energized sleep mode, as a result of which the circuit separation is canceled. There is thus a direct open-flow connection between the leak of the rear wheel brake **8d** and the first partial chamber **10** via the master brake cylinder **2**. The first partial chamber **10** runs empty. This can be seen in the graph of FIG. 2, which shows that the volume of the first partial chamber **10** rapidly drops to zero. During subsequent operation of the vehicle, the brake system is re-initialized and the empty first partial chamber **10** is detected. To enable braking operation, brake fluid is pumped out of the second partial chamber **11** into the first partial chamber **10**. This can be seen in FIGS. 2 and 3, which show how the second partial chamber



volume is reduced exactly to the extent that the volume of the first partial chamber **10** rises again. These steps are repeated during each subsequent initialization of the brake system, as a result of which the volume in the second partial chamber successively decreases. During the fourth initialization, the remaining brake fluid volume in the second partial chamber **11** is no longer sufficient to completely fill the first partial chamber **10**. Safe braking operation is then no longer possible.

(37) FIGS. **4** and **5** then show corresponding graphs for the method disclosed herein, which is shown step by step in FIG. **6**.

(38) In step **101**, a leak is detected in a wheel brake **8d** and at the same time a wheel brake **8c** without a leak is determined. During the transition to the sleep mode, the volume of the first partial chamber **10** is displaced into the second partial chamber **11**. For this purpose, in step **102**, the sequence valve **26** is opened and the circuit isolating valve **40** is closed. The inlet valve **6d** and the outlet valve **7d** of the wheel that has a leak are closed and the inlet valve **6c** and the outlet valve **7c** of the wheel without a leak are opened. In step **103**, the linear actuator **5** is then moved to its front end position, as a result of which the brake fluid volume contained therein is displaced via the inlet valve **6c** and the outlet valve **7c** of the wheel without a leak into the second partial chamber **11**.

(39) In step **104**, the circuit isolating valve **40** and the isolating valve **23** are opened. All the inlet valves **6** are closed. In step **105**, the linear actuator is then activated with a greatly reduced speed of 30-40 rpm to return to its rear end position. The resulting slight vacuum of about 0.1 bar draws only a small amount of brake fluid via the non-return valve **45** and out of the second partial chamber **11**. The linear actuator **5** draws brake fluid from the first partial chamber **10** through the open sequence valve **26**, circuit isolating valve **40**, isolating valve **23** and through the master brake cylinder **2**.

(40) FIG. **4** correspondingly shows, just like FIG. **2**, a steep drop in the volume of the first partial chamber **10** at the end of the active hour, but the volume is not lost owing to the leak and rather is fed to the second partial chamber **11**, as can be seen in FIG. **5**. There, the brake fluid volume is protected from leaking by the closed sequence valve **26**. When the brake system is subsequently initialized, the first partial chamber **10** is then refilled from the second partial chamber **11**.

(41) The steps of drawing brake fluid from the first partial chamber **10** and of displacing the brake fluid into the second partial chamber **11** are repeated in step **106** for a prespecified number of, for example three, cycles. As an alternative, a sensor can be provided in the first partial chamber **10**, and brake fluid is displaced until the sensor indicates that the first partial chamber **10** has been emptied.

(42) The embodiments prevent a loss of brake fluid in the sleep mode and thus ensures the brake system is able to function for a long time.

## Claims

1. A method for controlling a brake system comprising: transitioning a brake system to a standby state; detecting if a leak is present at one of the wheel brakes during the transition; and displacing a brake fluid volume from a first partial reservoir of a brake fluid vessel via a circuit isolating valve which is open into a second partial reservoir of the brake fluid vessel by activating an electric linear actuator of a the brake system.
2. The method as claimed in claim 1, further comprising separating the brake system into two partial circuits by closing the circuit isolating valve when the leak is detected, wherein a first partial circuit comprises at least a master brake cylinder with the first partial reservoir and also two first wheel brakes with associated inlet valves and outlet valves, and a second partial circuit comprises at least the linear actuator with the second partial reservoir and also two second wheel brakes with associated inlet valves and outlet valves.
3. The method as claimed in claim 2, wherein displacing the brake fluid volume from the first

partial chamber into the second partial chamber only occurs when the leak is present in the second partial circuit.

4. The method as claimed in claim 1, further comprising maintaining the circuit separation for a predetermined period of time during the transition to a standby state, and displacing the brake fluid volume of the first partial chamber into the second partial chamber at the end of the predetermined period of time.

5. The method as claimed in claim 1, wherein the first partial chamber is connected to the wheel brakes when the brake system is in a de-energized state.

6. The method as claimed in claim 1, further comprising displacing the brake fluid volume from the first partial reservoir into the second partial reservoir until the first partial reservoir is emptied.

7. The method as claimed in claim 1, further comprising: closing the circuit isolating valve to displace the brake fluid volume out of the linear actuator into the second partial reservoir; closing an inlet valve of the wheel brake with the leak is closed, the opening an inlet valve and an outlet valve of the wheel brake without the leak; and operating the linear actuator in a pressure build-up direction.

8. The method as claimed in claim 1, wherein displacing the brake fluid volume further comprising taking the brake fluid volume out of the first partial chamber into the linear actuator, and wherein an open-flow connection is established between the linear actuator and the first partial chamber and the linear actuator is operated in the pressure reduction direction.

9. The method as claimed in claim 8, further comprising establishing an open-flow connection via the master brake cylinder and/or via inlet valves and outlet valves of at least one wheel brake.

10. The method as claimed in claim 1, wherein displacing the brake fluid volume further comprises taking the brake fluid volume out of the first partial chamber, and wherein the linear actuator is operated such that a generated vacuum remains lower than a threshold value.

11. The method as claimed in claim 10, further comprising connecting the linear actuator to the second partial chamber via a non-return valve, which opens at a return differential pressure, and wherein the threshold value is lower than twice the return differential pressure.

12. A brake system for a motor vehicle comprising: a mechanically actuatable master brake cylinder a first partial reservoir of a brake fluid vessel fluidly connected to the master brake cylinder; a second partial reservoir of the brake fluid vessel; an electric linear actuator connected to the first and second partial reservoirs of the brake fluid vessel, wherein the master brake cylinder and the linear actuator are connected via an electrically closable circuit isolating valve; and a control unit of the brake system with instructions for: transitioning the brake system to a standby state; detecting if a leak is present at one of the wheel brakes during the transition; and displacing a brake fluid volume from the first partial reservoir via the circuit isolating valve which is open into the second partial reservoir by activating the electric linear actuator.

13. The brake system as claimed in claim 12, further comprising instructions for separating the brake system into two partial circuits by closing the circuit isolating valve when the leak is detected, wherein a first partial circuit comprises at least the master brake cylinder with the first partial reservoir and also two first wheel brakes with associated inlet valves and outlet valves, and a second partial circuit comprises at least the linear actuator with the second partial reservoir and also two second wheel brakes with associated inlet valves and outlet valves.

14. The brake system as claimed in claim 13, wherein displacing the brake fluid volume from the first partial chamber into the second partial chamber only occurs when the leak is present in the second partial circuit.

15. The brake system as claimed in claim 12, further comprising instructions for maintaining the circuit separation for a predetermined period of time during the transition to a standby state and displacing the brake fluid volume of the first partial chamber into the second partial chamber at the end of the predetermined period of time.

16. The brake system as claimed in claim 12, wherein the first partial chamber is connected to the

wheel brakes when the brake system is in a de-energized state.

17. The brake system as claimed in claim 12, further comprising instructions for displacing the brake fluid volume from the first partial reservoir into the second partial reservoir until the first partial reservoir is emptied.

18. The brake system as claimed in claim 12, further comprising instructions for: closing the circuit isolating valve to displace the brake fluid volume out of the linear actuator into the second partial reservoir; closing an inlet valve of the wheel brake with the leak is closed, the opening an inlet valve and an outlet valve of the wheel brake without the leak; and operating the linear actuator in a pressure build-up direction.

19. The brake system as claimed in claim 12, wherein displacing the brake fluid volume further comprising taking the brake fluid volume out of the first partial chamber into the linear actuator, and wherein an open-flow connection is established between the linear actuator and the first partial chamber and the linear actuator is operated in the pressure reduction direction.

20. The brake system as claimed in claim 12, further comprising instructions for establishing an open-flow connection via the master brake cylinder and/or via inlet valves and outlet valves of at least one wheel brake.

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